

Sales Data Analysis Report

Project Overview

This project analyzes a sales dataset using Python and Pandas to clean data, calculate sales metrics, and generate business insights.

Objectives

- Load CSV using Pandas
- Explore dataset
- Handle missing values
- Remove duplicates
- Calculate sales metrics
- Identify best-selling product
- Create analysis report

Dataset Information

- Dataset: sales_data.csv
- Rows: 100
- Columns: 5
- Columns: Product, Quantity, Price, Total_Sales, Region

Tools & Technologies

- Python
- Pandas
- VS Code
- Git & GitHub

Data Cleaning

- Checked missing values
- Filled missing values
- Removed duplicate records
- Verified data types

Exploratory Data Analysis

- Used head(), info(), describe(), shape, columns to understand the dataset

Sales Analysis

- Calculated Total Revenue
- Average Sales
- Highest Sale

- Lowest Sale
- Best-Selling Product

Results

- Replace with your actual program output:

Total Revenue: ₹12,365,048.00

Average Sales: ₹123,650.48

Highest Sale : ₹373,932.00

Lowest Sale : ₹6,540.00

Best Product : Laptop

Business Insights.

- Total business revenue was calculated successfully.
- The best-selling product generated the highest revenue.
- Average sales indicate overall business performance.
- No duplicate records affected analysis.
- Dataset was cleaned before calculations.

Conclusion

This project demonstrates the complete workflow of loading, cleaning, analyzing, and reporting sales data using Python and Pandas.

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Future Improvements

- Add charts
- Build Power BI dashboard
- Forecast future sales